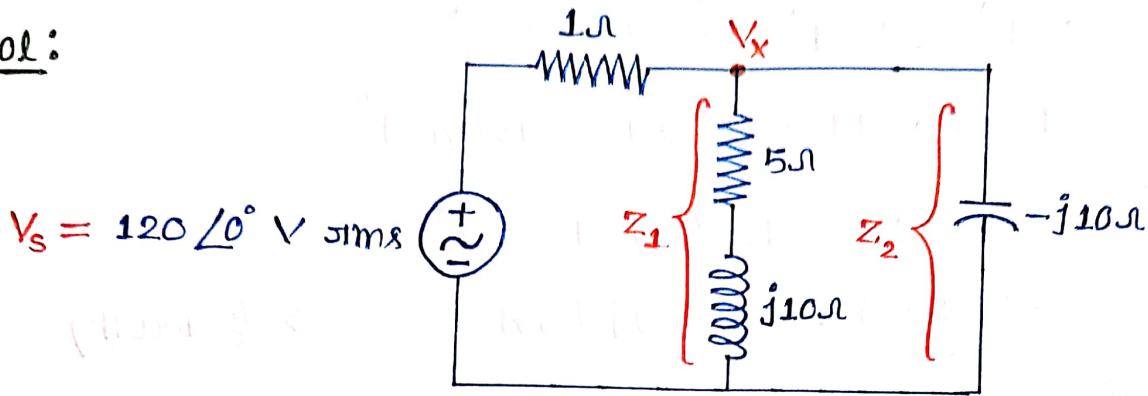


CTD QUIZ-2 SOLUTION (SET-A)

Sol:



$$V_x = \frac{(Z_1 \parallel Z_2)}{1 + (Z_1 \parallel Z_2)} V_s \quad \text{--- (1)}$$

$$Z_1 = (5 + j10) \Omega, \quad Z_2 = (-j10) \Omega$$

$$\therefore Z_1 \parallel Z_2 = \frac{(5 + j10)(-j10)}{(5 + j10 - j10)} = (20 - j10) \Omega$$

$$\begin{aligned} \text{Now, } V_x &= \frac{(20 - j10)}{1 + (20 - j10)} \times 120 \angle 0^\circ \\ &= (115.34 - 2.218j) \text{ V rms} \\ &= 115.36 \angle -1.1^\circ \text{ V rms} \end{aligned}$$

→ (1 Mark)

(a) Complex power absorbed by the 1 Ω resistor,

$$\begin{aligned} S_{1\Omega} &= \frac{1}{2} V \cdot I^* = V_{\text{eff}} \cdot I_{\text{eff}}^* = |V_{\text{eff}}|^2 / R \\ &= |V_s - V_x|^2 / R = \frac{|120 \angle 0^\circ - 115.36 \angle -1.1^\circ|^2}{1} \\ &= \frac{|5.16 \angle 25.41^\circ|^2}{1} = 26.63 \text{ VA} \end{aligned}$$

→ (2 Mark)

(b) Complex power absorbed by the -j10 Ω capacitor,

$$\begin{aligned} S_{-j10\Omega} &= V \cdot I^* = V_x \cdot \left(\frac{V_x}{-j10} \right)^* = \frac{|V_x|^2}{-j10} = \frac{(115.36)^2}{-j10} \\ &= -1330.87j \text{ VA} \end{aligned}$$

→ (2 Mark)



(c) complex power absorbed by the source,

$$\begin{aligned} S_{\text{source}} &= V \cdot I^* = V_S \cdot I_S^* = V_S \cdot \left(\frac{V_X - V_S}{1} \right)^* \\ &= 120 \angle 0^\circ \times (115.36 \angle -1.1^\circ - 120 \angle 0^\circ)^* \\ &= 120 \angle 0^\circ \times (5.16 \angle -154.59^\circ)^* \\ &= (-559.29 + 265.69j) \text{ VA} \end{aligned} \quad \rightarrow (2 \text{ MARK})$$

(d) Instantaneous power provided by the source,

$$p(t) = v_{\text{source}}(t) \times i_{\text{source}}(t) \quad \text{--- (2)} \quad \rightarrow (0.5 \text{ MARK})$$

$$v_{\text{source}}(t) = \sqrt{2} \times 120 \cos(\omega t + 0^\circ) \text{ V}$$

$$i_{\text{source}}(t) = \sqrt{2} \times \left(\frac{V_S - V_X}{1} \right)$$

$$= \sqrt{2} \times (120 \angle 0^\circ - 115.36 \angle -1.1^\circ)$$

$$= \sqrt{2} \times 5.16 \angle 25.41^\circ$$

$$= \sqrt{2} \times 5.16 \cos(\omega t + 25.41^\circ) \text{ A}$$

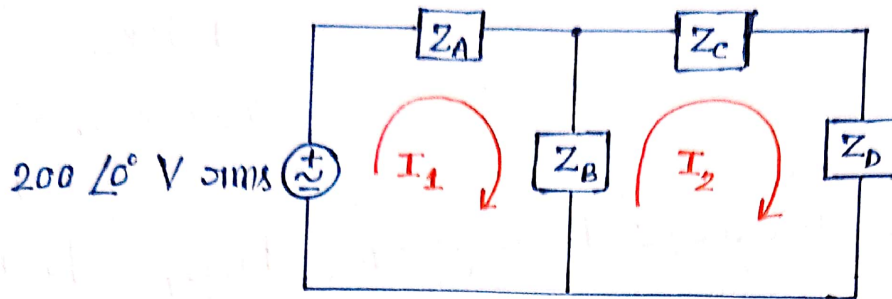
$\rightarrow (0.5 \text{ MARK})$

$$\begin{aligned} \therefore p(t) &= \sqrt{2} \times 120 \cos(\omega t + 0^\circ) \cdot \sqrt{2} \times 5.16 \cos(\omega t + 25.41^\circ) \\ &= 1238.16 \cos(\omega t) \cdot \cos(\omega t + 25.41^\circ) \text{ Watts} \end{aligned}$$

$\rightarrow (1 \text{ MARK})$

CTD QUIZ-2 SOLUTION (SET-B)

Sol:



Given- $Z_A = (5 - j2) \Omega$, $Z_B = 3 \Omega$, $Z_C = (8 + j4) \Omega$, $Z_D = 15 \angle -30^\circ \Omega = (12.99 - j7.5) \Omega$

By Mesh-1, $-200 \angle 0^\circ + (5 - j2)I_1 + 3(I_1 - I_2) = 0$ — (1)

By Mesh-2, $3(I_2 - I_1) + (8 + j4)I_2 + (15 \angle -30^\circ)I_2 = 0$ — (2)

By eqⁿ (1) & eqⁿ (2), we get —

$$I_1 = 25.3 \angle 15.06^\circ \text{ A rms}$$

$$I_2 = 3.13 \angle 23.36^\circ \text{ A rms}$$

} $\rightarrow (2 \times 1.5 \text{ MARK})$

(a)

Average power delivered to load $Z_A = P_{avg Z_A} = |I_1|^2 \times 5$
 $= (25.3)^2 \times 5 \rightarrow (0.5 \text{ MARK})$
 $= 3200.45 \text{ Watt}$
 $\rightarrow (1 \text{ MARK})$

Average power delivered to load $Z_B = P_{avg Z_B} = |I_1 - I_2|^2 \times 3$
 $\rightarrow (0.5 \text{ MARK})$
 $= (25.3 - 3.13)^2 \times 3$
 $= 1474.53 \text{ Watt}$
 $\rightarrow (1 \text{ MARK})$

Average power delivered to load $z_c = P_{avg z_c} = |I_2|^2 \times 8$
 $\rightarrow (0.5 \text{ Mark})$
 $= (3.13)^2 \times 8$
 $= 78.38 \text{ Watt}$

Average power delivered to load $z_D = P_{avg z_D} = |I_2|^2 \times 12.99$
 $\rightarrow (1 \text{ Mark})$
 $\rightarrow (0.5 \text{ Mark})$
 $= (3.13)^2 \times 12.99$
 $= 127.26 \text{ Watt}$
 $\rightarrow (1 \text{ Mark})$

(b) $PF_{source} = \cos(\theta - \phi)$
 $\rightarrow (0.5 \text{ Mark})$
 $= \cos(0^\circ - 15.06^\circ)$
 $= 0.9657 \text{ leading}$
 $\rightarrow (0.5 \text{ Mark})$